

581 Report- Auditing LLM Summaries

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1 Abstract

As Large Language Models (LLMs) are increasingly integrated into professional and academic workflows for automated summarization, the veracity of their outputs remains a critical concern, particularly in nuanced linguistic domains. This project investigates the reliability of LLM-generated summaries within the context of argumentative discourse, utilizing a corpus derived from the Reddit subreddit r/ChangeMyView. We evaluate the capacity of Gemini 3.0 Fast to accurately capture multi-party debates, identify shifts in participant perspectives, and maintain factual integrity across complex rhetorical structures. By comparing LLM-generated summaries against a manually annotated gold standard, we aim to quantify the limitations of current models in processing arguments and logical rebuttals in natural language.

2 Introduction

The rapid proliferation of Large Language Models has transformed document processing, with automated summarization serving as a primary use case for increasing workplace efficiency. However, the "black box" nature of these models introduces significant risks when users over-rely on summaries without verifying the underlying source material. This risk is amplified in argumentative contexts where the summary must not only compress information but also preserve the logical flow, identify conflicting stances, and detect subtle shifts in human persuasion. While existing evaluation metrics often focus on lexical overlap or fluency, they frequently fail to capture the semantic depth and "reasoning fidelity" required for high-stakes decision-making.

Our study addresses this gap by focusing on argumentative threads from the 'Change My View' community. This domain presents a unique challenge for AI; it requires a high degree of social rea-

soning to distinguish between the Original Poster (OP) and various challengers, and to pinpoint the exact moment of cognitive change—or the lack thereof. In this report, we present the preliminary findings of our week 5 evaluation, focusing on our custom annotation schema—Success, Accuracy, and Brevity—and our methodology for establishing inter-annotator agreement. We argue that assessing AI through the lens of argumentative text is essential for building more robust, trustworthy systems that can navigate the complexities of human disagreement.

3 Data

3.1 Corpus Acquisition and Composition

Our dataset consists of 400 data entry points sourced from the Reddit community r/ChangeMyView (CMV). We initially scraped 100 points and supplemented these with an additional 300 entries to form a robust corpus for evaluating argumentative summarization. Each entry represents an exchange where an Original Poster (OP) presents an argument for a potentially contentious point, followed by a back-and-forth argumentative thread where a different user attempts to change the OP's mind. A unique feature of this corpus is its paired structure: for every discussion, we include two branches—one where the OP acknowledges their viewpoint was changed some degree, and one where it was not.

3.2 Annotation Schema

To evaluate the Gemini 3.0 Fast summaries, four human annotators rated each output across three dimensions:

- **Sentiment (Success):** A boolean label (0 or 1) indicating if the model correctly identified whether the OP's mind was changed.

- **Accuracy:** A subjective 0–2 scale (0 = not accurate, 1 = moderately accurate, 2 = very accurate) assessing how well the model represented the core arguments.
- **Brevity:** A 0–2 scale (0 = few points, 1 = some points, 2 = most points) evaluating the model’s ability to capture the essential "key points" of the thread.

To ensure scientific validity, we calculated inter-annotator agreement using Krippendorff’s Alpha. This metric was chosen for its ability to handle multiple annotators and its support for ordinal data, which is necessary for our 0–2 weighted scales where a disagreement between 0 and 2 is more significant than one between 1 and 2.

3.3 Reasoned Data Split

To prepare for baseline model training, we implemented a "reasoned split" rather than a random 80-10-10 shuffle. To prevent data leakage, we split the data by thread number rather than by row. This ensures that the successful and unsuccessful branches of the same discussion remain in the same set, preventing the model from seeing the shared context of a test thread during training.

Furthermore, we stratified the split by reviewer id to maintain a balanced mix of human perspectives across sets. The final distribution resulted in 320 rows for training, 40 for development, and 40 for testing. While the dataset exhibits a natural class imbalance—particularly in the accuracy label, where class 2 is most frequent—this distribution was maintained across splits to reflect a realistic evaluation environment.

4 Methodology

The primary objective of this project phase is to establish a robust baseline for predicting the human-assigned Accuracy score (0, 1, or 2) of summaries generated by the Gemini 3.0 Fast model. By treating this as a supervised classification task, we aim to determine the extent to which traditional and simple neural architectures can mirror human judgment of "reasoning fidelity" in argumentative discourse.

4.1 Experimental Setup and Data Leakage Prevention

To ensure the integrity of our evaluation, we implemented a thread-aware data split. Because our

corpus contains paired "successful" and "unsuccessful" branches of the same Reddit thread, a naive random split would likely result in data leakage, where a model encounters the same underlying context in both the training and test sets.

To mitigate this, we partitioned the data by thread number. We utilized a stratified 80-10-10 split, ensuring that all variants of a single discussion remain within the same partition. Additionally, we stratified by reviewer id to prevent any single annotator’s subjective bias from disproportionately influencing a specific split.

4.2 Feature Representation

For both baseline models, we constructed a unified text representation by concatenating the original source argument with its corresponding AI-generated summary, prefixed by "SOURCE:" and "SUMMARY:" tokens respectively. This combined input allows the models to learn semantic relationships between the source premises and the model’s compression of those premises. Text features were extracted using a TF-IDF Vectorizer with the following parameters:

- **N-grams:** Unigrams and bigrams (1, 2) to capture local context.
- **Filtering:** Removal of English stop words and a minimum document frequency of 2.
- **Vocabulary:** Limited to the top 5,000 most frequent features.

4.3 Baseline Models

We developed two distinct pipelines to benchmark performance on the Accuracy label:

- **Traditional Baseline (Logistic Regression):** A standard linear model was selected for its interpretability and efficacy on small datasets. To address the natural class imbalance—where "high accuracy" (class 2) is over-represented—we utilized balanced class weights to penalize the misclassification of the minority classes (0 and 1).
- **Neural Baseline (MLP):** To explore non-linear patterns, we implemented a Multi-Layer Perceptron (MLP). Given the high dimensionality of the TF-IDF vectors, we applied Truncated SVD to reduce the feature space to 100 components. The architecture

consists of a single hidden layer with 64 neurons, utilizing early stopping to prevent overfitting on our low-resource dataset.

Both models are evaluated on the development set using Macro F1-score as the primary metric, as it provides a more conservative and equitable assessment of performance across imbalanced classes than raw accuracy.

4.4 Ensemble Methods

To leverage the complementary strengths of the traditional and neural pipelines, we implemented a soft-voting ensemble. Unlike hard voting, which only considers final class predictions, soft voting aggregates the predicted probabilities from each baseline’s predict_proba function, allowing the ensemble to account for the models’ relative confidence levels. Recognizing that the traditional logistic regression model exhibited a stronger Macro F1 performance in initial trials, we developed a "motivated ensemble" by applying a manual weight search. We found that a weighted distribution—assigning 75

4.5 Transfer Learning and Data Augmentation

In the second stage of our experimental pipeline, we introduced a secondary source of information to supplement the sparse TF-IDF features used in Sprint 1. We integrated pretrained GloVe word embeddings (glove-wiki-gigaword-50) to provide the models with a dense semantic representation learned from external, large-scale corpora. Given the small size of our annotated dataset, this transfer learning approach serves as a critical regularizer, allowing the model to capture lexical relationships that TF-IDF alone might overlook. We generated separate average embeddings for the source argument and the generated summary, then concatenated these vectors to maintain a clear distinction between the input context and the model’s output. For the traditional model (B1+T), these features were appended to the TF-IDF vectors, while for the neural model (B2+T), the embeddings were combined with the SVD-reduced features before being passed to the hidden layers of the MLP.